

10.5 How our energy is produced

Energy is central to modern-day living. It powers our machinery, heating and cooling, food production, and many more processes that form part of our daily life. The energy that we use comes from two types of energy sources: **non-renewable** and **renewable**. Understanding the difference in these sources and their impacts on humans, the economy, and the planet is important when considering the long-term sustainability of energy production.

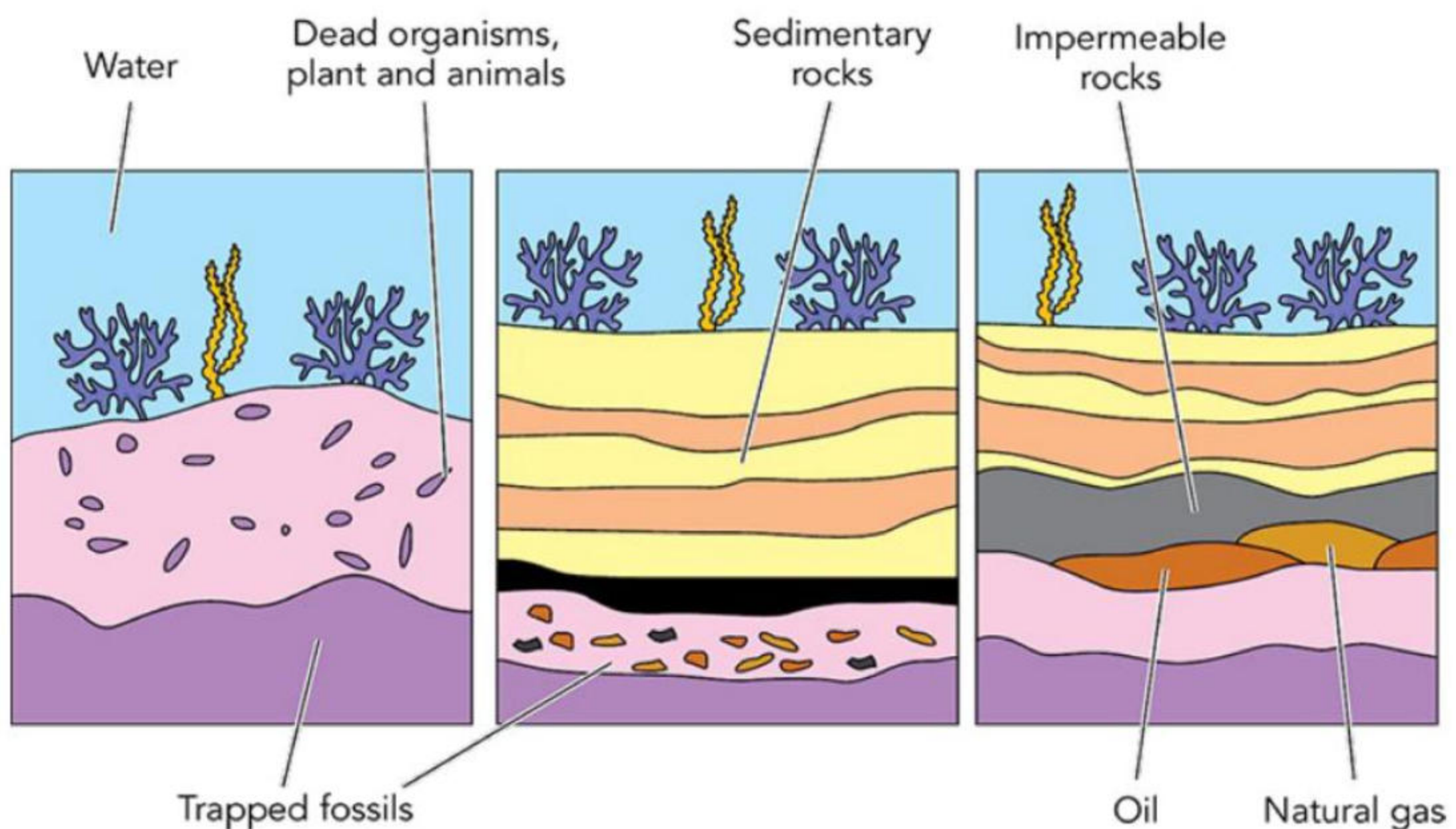
Non-renewable energy sources

Non-renewable energy is produced from resources that are finite. This means they will not be replenished for hundreds of thousands, if not millions of years. These resources include the fossil fuels oil, coal, and natural gas, and uranium that is used for the production of nuclear energy.

Oil and gas

The process that creates oil and gas occurs under the ocean, over millions of years (Source A). Dead plankton, algae, and bacteria sink to the ocean floor where they mix with clay-like materials. When this mix of clay and organic matter occurs in still-water environments with little or no oxygen, it settles on the ocean floor rather than decaying. Over time, this buried material becomes sedimentary rock. When buried between 2 km and 4 km deep, the pressure and temperature transform the rock into oil and natural gas.

Oil is formed when temperatures reach between 90°C and 160°C. Natural gas forms at higher temperatures. Oil and gas are lighter than water, so they rise through the pores in the rocks. If they reach an impermeable layer of rock or clay-like material, the oil and gas can no longer move and collect here.

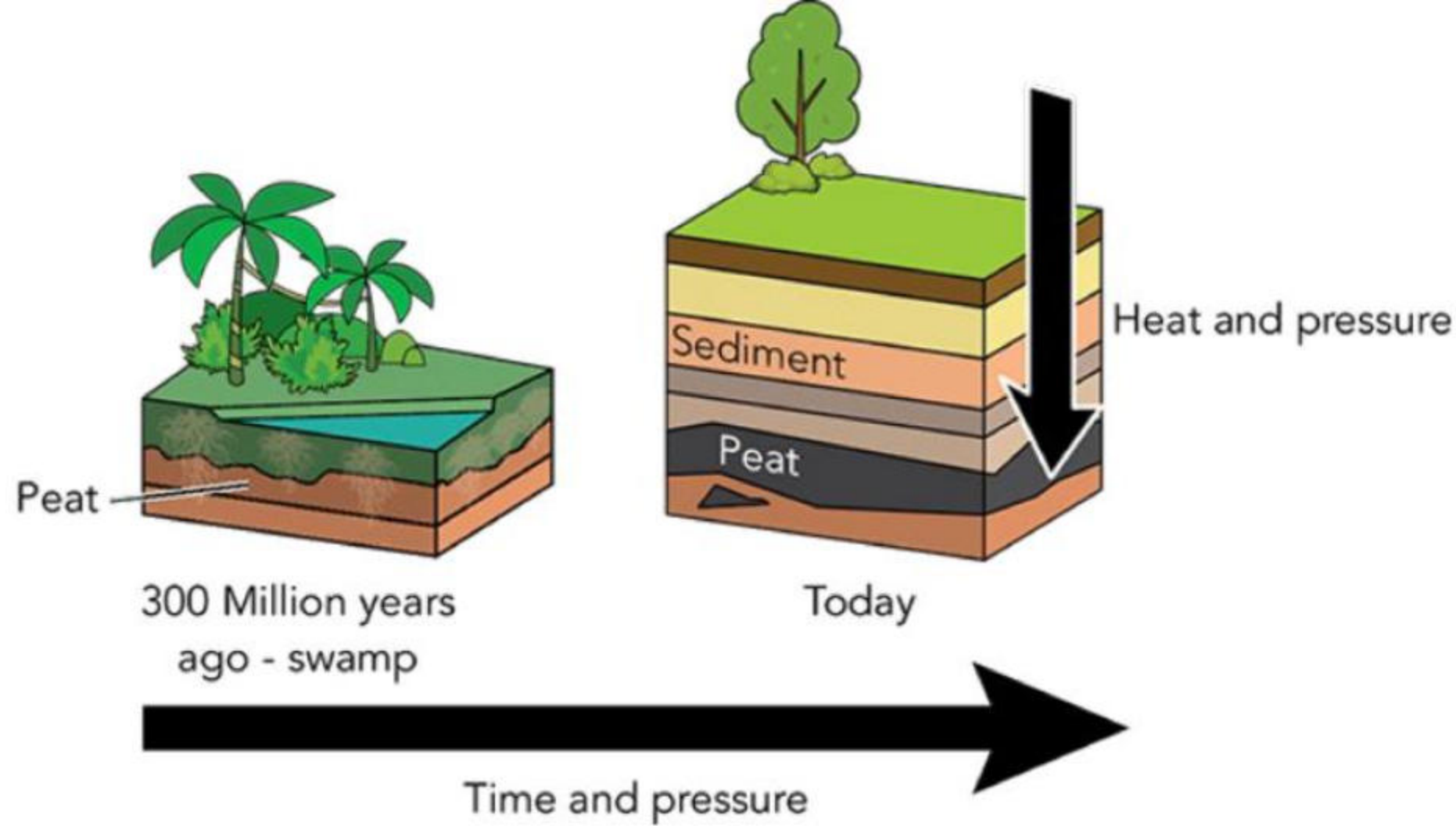


Source A: The formation of oil and natural gas at the bottom of the ocean

Coal

Millions of years ago, the earth was covered with trees and plants. These plants died in swamps and along riverbanks where soils and mud covered them. As they sunk into the swamps and riverbed, they decomposed into peat. Over time, the pressure of the ground above and the high temperatures (50°C to 150°C) converted the plant material into coal. Coal is an energy-rich, carbon-dense, black sedimentary rock.

Coal travels closer to the surface as plants bend and fold the earth's crust, or the surface material erodes.



Source B: The formation of coal from plant material over millions of years

Nuclear power

Nuclear power is a non-renewable energy resource because it uses uranium, a radioactive-heavy metal, which has to be mined from the earth. At the current rate of consumption, the accessible uranium supply will last approximately 230 years before we run out. However, the amount of energy that a small amount of uranium produces is significant. This makes it a more sustainable energy supply than coal and gas.

TASK 1: Study Sources A B

a Use a table to compare the formation of oil and gas with coal. Look for both similarities and differences in their formation.

	Oil and gas	Coal
Time		
Original organic material		
Where it forms		
How it forms		

b With reference to Sources A and B, explain why nuclear energy is considered a more sustainable energy source.

Renewable energy sources

Renewable energy uses sustainable energy resources such as wind power, solar power, and moving water to generate energy that is converted into electricity. Unlike non-renewable energy sources, renewable energy sources will not run out. They can be replenished quickly. Currently only a small percentage of global energy production comes from renewable resources.

Most electricity-generating systems rely on turbines to generate power. Wind and water flow can easily turn turbines directly. Heating water to create steam can also turn turbines. This heating is done by burning fossil fuels or by using other heat sources. Only one energy source, solar power, does not rely on turbines for generating electricity. Instead, an electrical charge in the solar cell generates electricity.

Figure 10.7: Renewable energy sources

Tidal and wave energy: uses the power from the movement of waves and the tides in river mouths to turn turbines and generate electricity.

Biomass: energy obtained from organic materials such as plant and animal waste

Wind energy: captures the movement of air to turn the blades on wind turbines. This turning motion is used to generate electricity.

Hydro-electric power (HEP): electricity is produced using moving water. A dam is built across a river, blocking the flow of water. This water is then channeled under pressure through pipes to turn turbines which generate power.

Renewable energy sources

Solar energy: produced by harnessing the power of the sun. The energy from the sun is converted into electrical energy through solar panels. The solar panels use minerals that react to sunlight, generating a charge which is captured as energy.

Geothermal energy: uses heat from within the Earth's crust to heat water or generate electricity. Water is pumped down into the earth and returns to the surface as steam. This steam is used to turn turbines which generate electricity. Water can be heated by being pumped underground. It then returns to the surface heated. This can be used to heat homes or as a source of hot water.

► Image Description

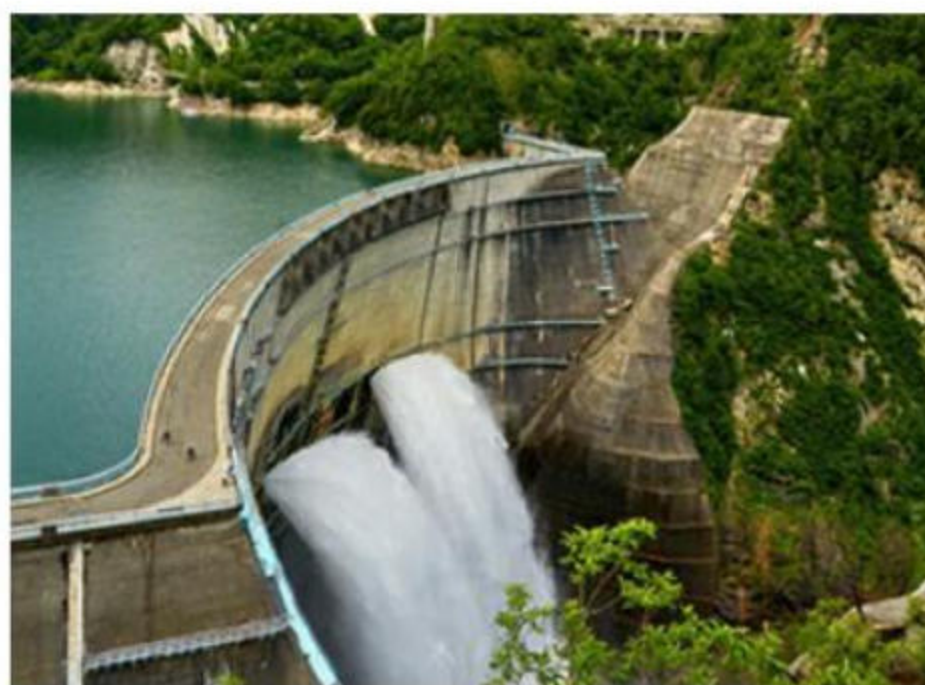
We can further divide the energy generated from biomass into **fuelwood** and **biogas**.

- a** Fuelwood is wood that we grow and use to generate heat and light. In many parts of the world, people use fuelwood for cooking. It relies on trees as a source for the wood. If humans harvest trees at a faster rate than they can regrow, then this renewable resource can become a non-renewable energy source.
- b** Biogas is generated when plant or animal waste ferments to create methane. We can then use this gas as an energy source for activities like cooking, or to fuel vehicles or heat homes.

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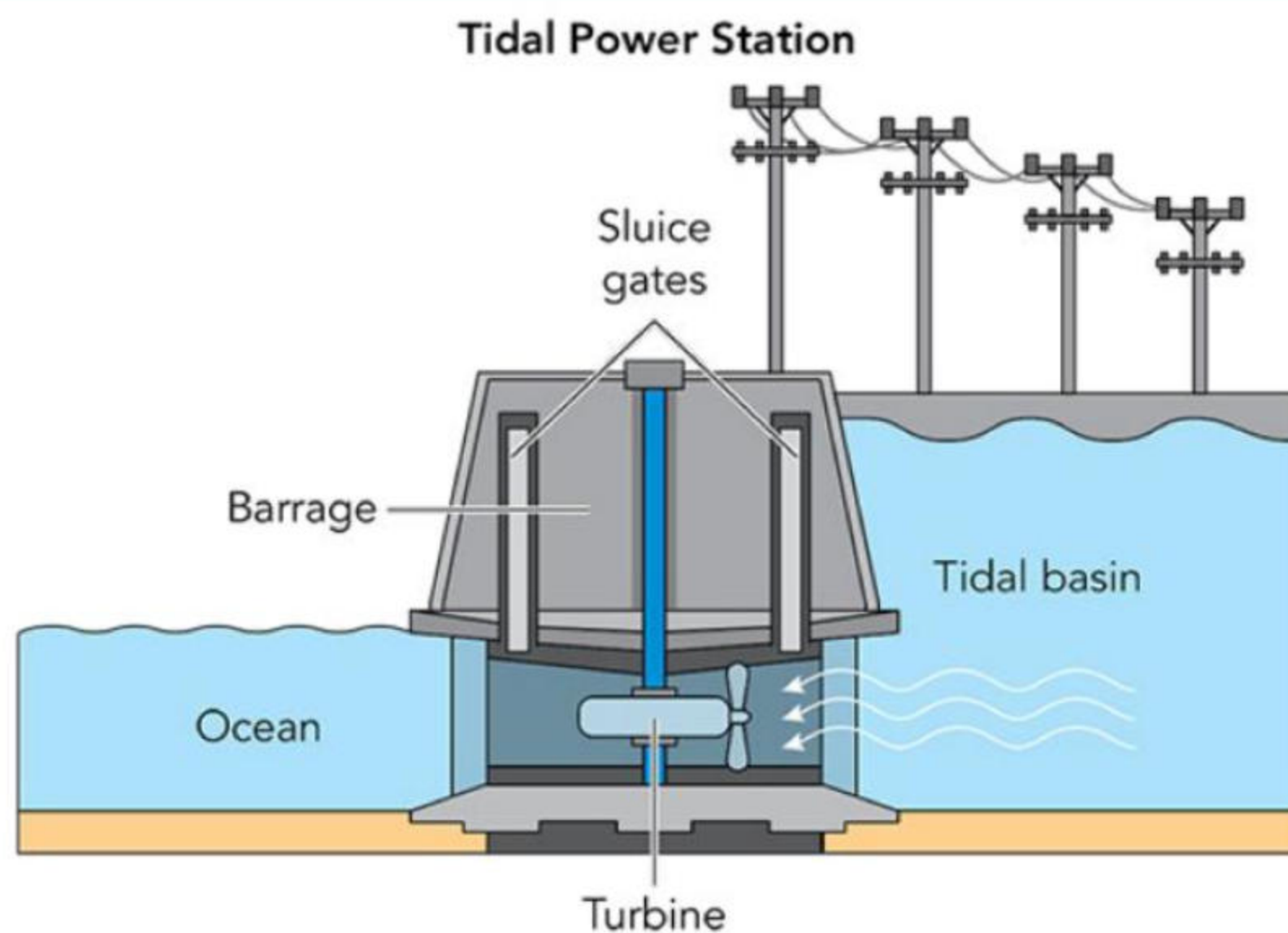


Source **C**: Wind turbines and solar panels on an energy farm in the Netherlands



Source **D**: Flowing water over the hydro-electric dam of Kurobe in Japan





Source **F**: Tidal energy produced by underwater turbines which are powered by the moving currents

TASK 2: Study Sources **C** - **F**

- a** Describe the similarities between generating tidal and wind energy.
- b** Solar power does not require turbines to generate electricity. In pairs, discuss how you think this makes solar energy more suitable for remote locations.
- c** With reference to Source F, describe how tidal energy can generate electricity.
- d** Explain why we can consider fuelwood both a renewable and non-renewable energy source.

REFLECTION

How did you find making a comparison between two seemingly different systems in Task 2a (wind and tidal energy)? Did you take time to consider how the electricity is generated to look for those similarities? If not, read through how electricity is generated again to help you.